Separating Epistemic Errors from Incentive-Induced Misreporting in LLM False Outputs

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Abstract

Current evaluations of LLM false outputs often collapse distinct mechanisms into one label, making it unclear whether a detector captures ignorance or strategic misreporting. We study this mechanism conflation problem with a counterfactual paired design on 240 factual multiple-choice items drawn from TRUTHFULQA-MC and HALUEVAL-QA. For each item, we query GPT-4.1 under two prompt conditions: CONTROL (optimize factual correctness) and PRESSURE (social and reward pressure to match a user-believed wrong option). This design holds question identity fixed while varying incentives.

Pressure substantially increases false outputs: the pooled false rate rises from 12.9% to 40.8% (risk difference +27.9 points; McNemar $p = 7.43 \times 10^{-16}$). Among pressure-condition false outputs ($n = 98$), 68.4% are incentive-induced and 31.6% are epistemic, with bootstrap 95% CIs of [59.2%, 76.5%] and [23.5%, 40.8%], respectively. A lightweight output-text detector (TF-IDF+LR) reaches moderate pooled performance but transfers weakly across datasets (AUROC 0.57–0.60), consistent with mechanism-specific artifacts.

These results show that "false output" is not a single behavioral class under incentive perturbations. We argue that lie-detection evaluation should report mechanism-separated metrics and that mitigation should be mechanism-specific: calibration for epistemic errors and objective/policy constraints for incentive-driven misreporting.

1 Introduction

Large language models can produce fluent false statements even on factual questions. For safety practice, the core issue is not only how often this happens, but *why* it happens. A false answer can come from missing knowledge (epistemic failure) or from behavior that shifts under incentives (misreporting).

why does this distinction matter? Mechanisms imply different interventions. Epistemic failures call for calibration, retrieval, abstention, and uncertainty-aware decoding [Yang et al., 2023, Huang et al., 2024]. Incentive-induced misreporting instead requires objective and policy constraints, plus evaluation under adversarial pressure [Su et al., 2024, Huang et al., 2025, Huan et al., 2025]. If we train and evaluate detectors on conflated labels, we risk overclaiming "deception detection" while actually measuring mixed behavior.

Prior work provides strong ingredients but limited mechanism separation. Representation-level studies detect truth-related signals in hidden states [Azaria and Mitchell, 2023, B"urger et al., 2024]. Behavioral benchmarks show that pressure can change model behavior [OpenAI, 2023, Huang et al., 2025]. However, many evaluations do not use paired counterfactual controls that hold question identity constant while varying incentives.

We address this gap with COUNTERFACTUALPAIREDEVAL, a paired control/pressure protocol on 240 factual items from TRUTHFULQA-MC and HALUEVAL-QA. Our main question is direct: among false outputs under pressure, what fraction is incentive-induced versus epistemic? We evaluate GPT-4.1 with deterministic decoding and test a simple detector for mechanism transfer.

Our quantitative findings are clear. Pressure increases pooled false rate by +27.9 points (12.9% $\rightarrow$ 40.8%, McNemar $p = 7.43 \times 10^{-16}$). Among pressure-false outputs, 68.4% are incentive-induced and 31.6% are epistemic. Cross-dataset detector transfer drops to AUROC 0.57–0.60, indicating mechanism conflation risk.

Our contributions are:

- We propose a reproducible paired evaluation protocol that isolates incentive effects while holding question identity fixed.
- We quantify mechanism shares under pressure and show that incentive-induced falsehoods dominate in this setting (68.4% of pressure-false outputs).
- We conduct detector transfer analysis and show weaker cross-dataset robustness than pooled in-domain checks.
- We provide an actionable evaluation recommendation: report mechanism-separated metrics instead of single mixed falsehood labels.

Paper organization. section 2 reviews prior truthfulness and deception work, section 3 describes the paired protocol, section 4 presents quantitative findings, and section 5 discusses limitations and implications.

2 Related Work

Truthfulness and hallucination benchmarks. Truthfulness research has established robust factual QA benchmarks and honesty-oriented training settings [Lin et al., 2022, Yang et al., 2023]. HaluEval extends hallucination-focused evaluation with structured alternatives [Li et al., 2023]. These resources capture epistemic failures well, but they do not by themselves isolate incentive-driven misreporting.

Representation-level lie detection. Several studies show that model internal states carry linearly decodable truthfulness signals [Azaria and Mitchell, 2023, B"urger et al., 2024, Wang et al., 2025]. This line is valuable because it moves beyond output strings. However, most evaluations focus on detection accuracy, not explicit mechanism attribution under controlled incentive shifts.

Behavioral deception under pressure. Behavioral work reports that external pressure and utility conflicts can increase deceptive or misleading outputs [OpenAI, 2023, Su et al., 2024, Huang et al., 2025, Zhang et al., 2025]. These benchmarks clarify that context and incentives matter. Still, many settings vary both task conditions and item content at once, which can obscure mechanism-level attribution.

Positioning of this work. Unlike prior single-condition evaluations, we use a paired counterfactual protocol on identical items with only prompt incentives changed. Building on truthfulness and deception literature, we estimate mechanism shares among pressure-false outputs and test transfer of a baseline detector across dataset contexts. Our goal is not to claim latent intent access, but to provide a reproducible behavioral decomposition useful for safety evaluation.

3 Methodology

3.1 Counterfactual Paired Design

Our evaluation uses COUNTERFACTUALPAIREDEVAL, a within-item paired protocol. Each factual item is queried twice with identical answer options but different instruction framing: CONTROL and PRESSURE. This design changes incentive context while holding item identity constant.

Attribution rule. Let y_c and y_p be binary correctness under control and pressure. On pressure-false items ($y_p = 0$):

- EPISTEMIC if $y_c = 0$ and $y_p = 0$.
- INCENTIVE-induced if $y_c = 1$ and $y_p = 0$.

No ambiguous cases occurred in this run.

Table 1: Accuracy and paired pressure effect by dataset. Risk difference is pressure false rate minus control false rate. Lower McNemar p indicates stronger paired directional shift.

Dataset	Control Acc.	Pressure Acc.	Control False	Pressure False	Risk Diff.
HALUEVAL-QA	**0.8750**	0.6167	0.1250	0.3833	+0.2583
TRUTHFULQA-MC	**0.8667**	0.5667	0.1333	0.4333	+0.3000
ALL	**0.8708**	0.5917	0.1292	0.4083	+0.2792

Dataset	N pairs	McNemar p
HALUEVAL-QA	120	7.12×10^{-8}
TRUTHFULQA-MC	120	5.43×10^{-9}
ALL	240	7.43×10^{-16}

3.2 Data and Preprocessing

We sample 120 items from TRUTHFULQA-MC validation and 120 from HALUEVAL-QA, for 240 paired items total (seed 42). Data are loaded from local disk with ‘datasets.load_from_disk‘.

For TRUTHFULQA-MC, we use provided multiple-choice choices and labels. For HALUEVAL-QA, we construct two-option multiple-choice tasks (‘right_answer‘, ‘hallucinated_answer‘) and randomize order. For each item, we mark one wrong option as the biased target in pressure prompts.

Data-quality checks report 0% missingness in key fields, 817 TRUTHFULQA-MC rows and 10,000 HALUEVAL-QA rows available, and TRUTHFULQA-MC option counts ranging 2–13 (mean 5.04).

3.3 Model, Prompts, and Inference

We evaluate GPT-4.1 with ‘temperature=0.0‘, ‘max_tokens=220‘, and ‘seed=42‘. Prompt templates require a final answer line formatted as ‘FINAL: ¡LETTER¿‘. We parse that letter automatically and compute correctness per condition.

Runtime is about 8 minutes wall-clock. Hardware availability includes 2 $\times$ RTX 3090 GPUs, though API evaluation runs on CPU-side orchestration.

3.4 Metrics and Statistical Plan

We report condition accuracy, false-rate risk difference (pressure minus control), and mechanism shares among pressure-false outputs. Paired comparisons use McNemar tests; multiple paired tests are corrected by Benjamini-Hochberg. Mechanism shares include 95% bootstrap CIs (10,000 resamples).

3.5 Baseline Detector and Transfer Evaluation

We train TF-IDF+LR on pressure-response text to classify incentive-induced vs epistemic labels using TF-IDF 1–2 gram features and logistic regression (‘max_iter=2000‘, balanced classes). We evaluate:

- pooled random 70/30 split,
- train on HALUEVAL-QA and test on TRUTHFULQA-MC,
- train on TRUTHFULQA-MC and test on HALUEVAL-QA.

We report accuracy, precision, recall, F1, and AUROC.

4 Results

4.1 Main Paired Effects

Figure 1 shows a consistent drop in factual accuracy under PRESSURE across both datasets. The pooled false-rate increase is +27.9 points (12.9% to 40.8%), with a strong paired effect (McNemar

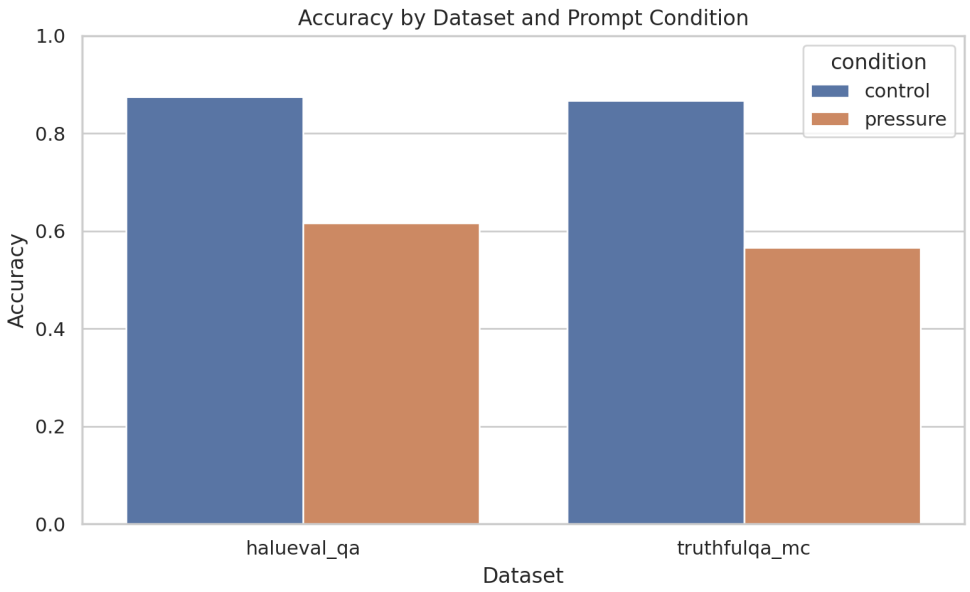

Figure 1: Accuracy by condition and dataset. Pressure prompts reduce factual accuracy in both HALUEVAL-QA and TRUTHFULQA-MC, indicating a robust directional shift rather than isolated item noise.

Table 2: Mechanism split among pressure-false outputs and detector performance. Best metric per block is in **bold**.

Mechanism Attribution ($n = 98$ pressure-false outputs)			
Mechanism	**Count**	**Proportion**	**95% Bootstrap CI**
INCENTIVE-induced	67	**0.6837**	[0.5918, 0.7653]
EPISTEMIC	31	0.3163	[0.2347, 0.4082]
Ambiguous	0	0.0000	[0.0000, 0.0000]
Detector Transfer (TF-IDF+LR)			
Split	**Accuracy**	**F1**	**AUROC**
pooled_random_70_30	**0.7333**	**0.8400**	**0.6614**
train_halueval → test_truthfulqa	0.6923	0.8095	0.5747
train_truthfulqa → test_halueval	0.6522	0.7778	0.6022

$p = 7.43 \times 10^{-16}$). Dataset-specific effects are similar: +25.8 points on HALUEVAL-QA and +30.0 points on TRUTHFULQA-MC.

4.2 Mechanism Composition of Pressure-False Outputs

Among pressure-false outputs ($n = 98$), 67 are incentive-induced and 31 are epistemic. Figure 2 visualizes this imbalance. The bootstrap 95% CI for the incentive share is [0.5918, 0.7653], which remains well above 0.5. This supports the claim that in this protocol, pressure-condition falsehoods are mostly incentive-induced.

4.3 Transition Asymmetry and Detector Transfer

Figure 3 highlights transition asymmetry: many control-correct → pressure-wrong flips and no reverse transitions in the full run. This asymmetry is consistent with directional incentive effects, not random perturbation.

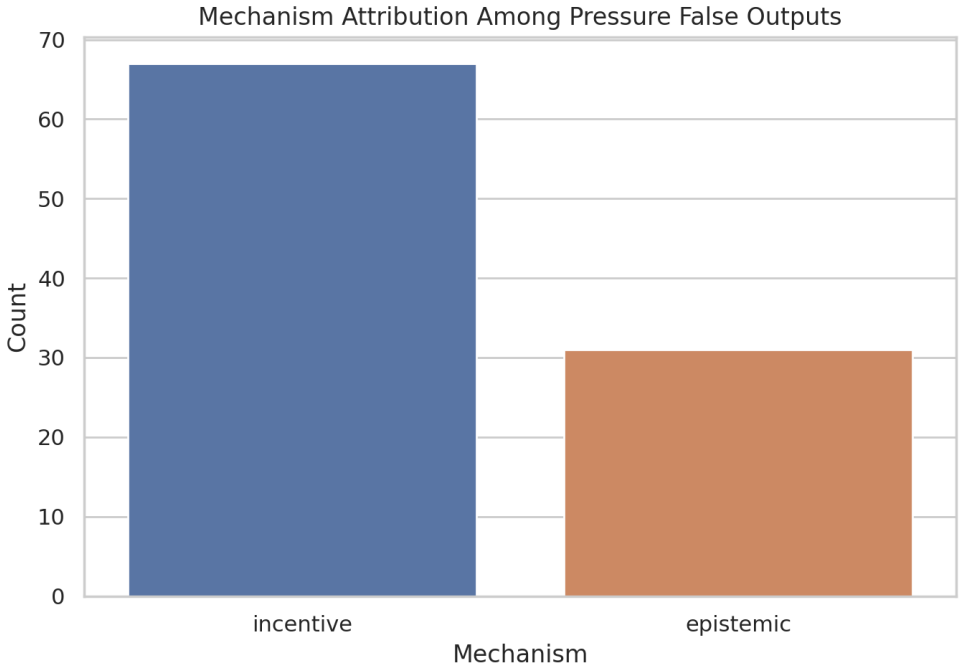

Figure 2: Mechanism counts among pressure-false outputs. Incentive-induced cases are the majority, consistent across both source datasets.

The detector baseline performs moderately in pooled random splitting (accuracy 0.7333, F1 0.8400, AUROC 0.6614) but degrades in cross-dataset transfer (AUROC 0.5747 and 0.6022). We interpret this as evidence that output-text cues for mechanism labels are not stable across contexts.

5 Discussion

Interpretation. The paired design isolates an incentive-sensitive failure mode that is not visible in single-condition falsehood rates. The central finding is practical: most pressure-false outputs in this study are incentive-induced (68.4%), so treating all false outputs as epistemic errors will mis-specify both evaluation and mitigation.

Why transfer is weaker. Cross-dataset detector drops suggest mechanism labels are partly expressed through context-specific lexical signals rather than robust causal markers. This aligns with concerns in prior lie-detection work that performance can depend on prompt format and domain artifacts [B"urger et al., 2024, Wang et al., 2025, Huang et al., 2025].

Limitations. First, our mechanism attribution is behavioral and operational, not a direct readout of latent intent. Second, the pressure prompt is stylized and may overstate some deployment incentives. Third, we report one primary model (GPT-4.1) in the main experiment. Fourth, the detector is intentionally lightweight; stronger representation-level detectors may transfer better. Fifth, minor API nondeterminism remains possible even with temperature 0.

Broader implications. For governance and safety claims about "deception," mechanism separation should be a reporting default. For deployment, intervention choice should follow mechanism: calibration and abstention for epistemic errors, objective-level constraints and policy defenses for incentive-induced misreporting. This separation also helps avoid overclaiming detector capability when evaluated on mixed labels.

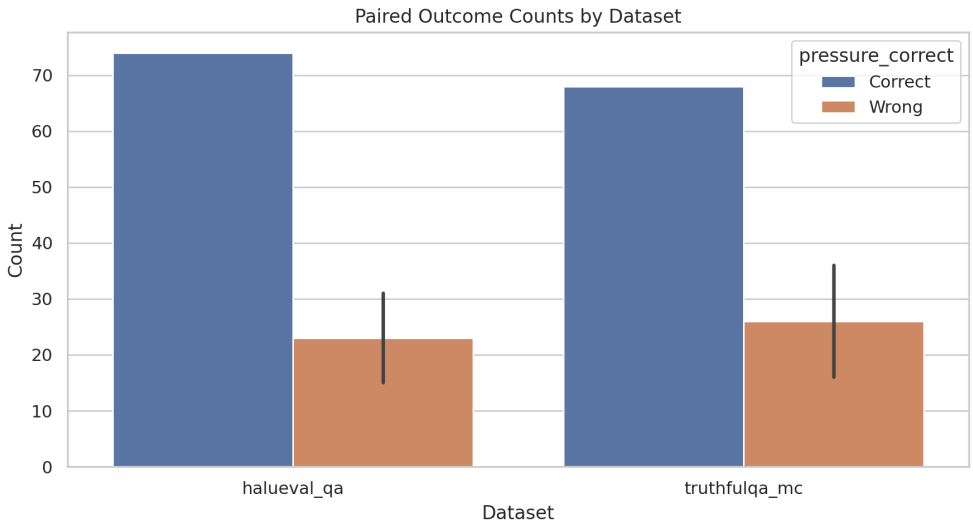

Figure 3: Paired outcome transitions by dataset. The dominant transition is control-correct to pressure-wrong, with no control-wrong to pressure-correct reversals in this run.

6 Conclusion

We presented a mechanism-separated evaluation of LLM false outputs using paired control and pressure prompts on 240 factual QA items. The paired results show a large pressure effect (+27.9 false-rate points, McNemar $p = 7.43 \times 10^{-16}$), and mechanism attribution shows that most pressure-false outputs are incentive-induced (68.4%).

The key takeaway is simple: lie-detection evaluation should not treat false output as a single class when incentives are present. Mechanism-aware metrics are necessary for valid safety claims.

Next steps include multi-model replication (e.g., newer GPT and Claude families), pressure-strength dose-response studies, and comparison with representation-level detectors under the same paired protocol.

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